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Using the Internet of Things (IoT) in Building an Irrigation Control System

Hossein Mosaffa[1*] and Hajar Atrianfar[2]

[1]*Student, Amirkabir University of Technology (Tehran Polytechnic), h.mosaffa@hotmail.com*

[2]*Faculty member, Amirkabir University of Technology (Tehran Polytechnic), atrianfar@aut.ac.ir*

Abstract: The goal of this project is to use Internet of Things (IoT) technology in irrigating plants and optimizing water consumption. The value from a soil moisture sensor is sent to a web page through the MQTT protocol and displayed there, and based on those values, solenoid valves are opened or closed. To achieve this goal we used open-source tools together with the advanced Raspberry Pi and ESP32 WiFi modules. The Node-RED programming environment was also used to build the web page we wanted.

Keywords: *Irrigation system, Internet of Things (IoT), Raspberry Pi, ESP32, Node-RED*

1- Introduction

In recent years Iran has faced a water crisis. At the same time, 90 percent of water consumption is in the agriculture sector. Unfortunately, because irrigation planning and management are not fully carried out on farms, in orchards and in green spaces, irrigation efficiency is very low. With a smart irrigation system one can truly take control of timing in the irrigation process and of the amount of water used, and with correct, scientific and logical use of such a system we would no longer see water wasted, along with the problems that come with it, at a time when the country is facing water scarcity. Using the Internet of Things can be a way to control irrigation systems [2].

The Internet of Things is a system of physical objects that can be discovered, monitored, controlled and interacted with by electronic devices. These objects communicate with one another over various network interfaces and can even connect to the wider Internet. We owe the spread and progress of the Internet of Things to advances in embedded devices, which introduced a new class of objects called smart objects. A smart object is a physical object equipped with a sensor and an actuator. It has memory and can be programmed, and in addition it has a communication interface (wired or wireless) through which one can communicate with it [1].

Remote control of electronic and electrical devices was a known subject from 1990. With the birth of the Internet in 1989, the connection of objects took on a new form. The Trojan Room coffee pot in the Computer Laboratory of the University of Cambridge is one of the first such applications. A camera, using the video card of an Acorn Archimedes computer, showed the state of the coffee maker over the local network in 1991. In November 1993 the camera was connected to the Internet through a browser by Daniel Gordon and Martin Johnson. In this way an object was controlled over the Internet for the first time, and this was regarded as the beginning of IoT. At 9:54 on 22 August 2001 (UTC) this camera was switched off for good, and the computer department moved to a new location [3].

The IoT network has very great potential and is growing rapidly. Millions of devices connected to the Internet of Things are currently active around the world. It is predicted that by 2020 the number of these devices will reach 30 billion. Researchers also believe that the global value of the IoT market in that year will reach 7,100 billion dollars, a striking figure that shows the importance of the Internet of Things and its bright future. Imagining super-smart cities in the very near future, in which every object is connected to the Internet and able to make decisions, is by no means far-fetched given concepts such as the Internet of Things and cloud computing, and extensive studies are under way [7].

The general operation of the designed system is as follows. For each existing tree we have a soil moisture sensor, a WiFi module and a solenoid valve. The data from all these sensors is sent wirelessly, through all the WiFi modules, to a central unit, which is the Raspberry Pi mini-computer. A web page has been designed to display the data from all of these sensors, and it resides in the Raspberry Pi's memory. The Raspberry Pi places the data it receives from the sensors on the web page, and from anywhere we have Internet access we can see the sensor data through the IP address of that web page. In addition, a switch dedicated to each solenoid valve has been placed on the web page, and turning these switches off and on opens and closes the solenoid valves.

1-1 Raspberry Pi

Raspberry Pi is a popular series of single-board computers. Because a WiFi module is built into it, it has its own IP address and one can communicate with it over the Internet, and this is the reason for its popularity in the Internet of Things field. In this project the Raspberry Pi acts as the central information unit. The web page resides on it, and all exchanged data goes to it first and from there finds its correct way to the solenoid valves.

1-2 ESP32 WiFi Module

The ESP32 WiFi module is in fact the smart object we discussed in the introduction. It has digital and analog pins, and as a result the moisture sensor and the solenoid valve (actuator) are connected to it. It has memory, we can program it, and we can communicate with it using various communication protocols.

1-3 Other Components

We used a 12-volt solenoid valve as the actuator. To drive it and to connect it to the ESP32 WiFi module we need a relay, a diode, a resistor and a transistor.

1-4 The MQTT Protocol

To place the sensor data on the web page, and for communication between the web page and the solenoid valves, we used the popular MQTT protocol.

MQTT is a publish–subscribe protocol. This means that MQTT clients subscribe to a topic, and whenever a message on that topic is published, they receive the information. Publishers and subscribers do not talk to each other directly but through an intermediary called a broker. Our broker is in fact the Raspberry Pi: all information goes to it first, and from there it sends the message with the relevant topic to the clients connected to it.

1-5 Node-RED

Node-RED is a visual tool for the Internet of Things. It is an open-source project that supports several protocols but has a strong focus on web protocols, and it benefits from a large community of developers who create new modules almost every day. The goal of Node-RED is to make programming and building the web page very easy for us.

2- Implementation Process

2-1 Raspberry Pi Settings

To configure the Raspberry Pi, we first set it up (installing the operating system on an SD card and getting access to its terminal).

As we said in the introduction, the broker's job is to receive all messages, filter them, decide who has requested those messages, and publish each message to all of its subscribers.

We now want the Pi to play the role of the broker in the project. There are various brokers, and one of them is called Mosquitto Broker, which is installed on the Pi. Also, to build a web page that displays the sensor data and lets us turn the solenoid valve off and on, we must install it [Node-RED] on the Pi, and its dashboard must be installed as well.

2-2 ESP32 Settings and Solenoid Valve Setup

The ESP's tasks are to connect to WiFi and to communicate with the MQTT broker. Our broker is the Raspberry Pi, so all the ESPs connect to the Raspberry Pi. To make this happen we use the Arduino programming environment and program the modules.

In the modules' program we use the WiFi.h and PubSubClient libraries, which respectively connect the module to the Internet and establish MQTT communication with the Raspberry Pi and exchange information with it.

To control the solenoid valve we used one of the ESP32's digital pins. To achieve this we used several components: an NPN transistor (C945), a diode, a resistor, a 5-volt relay, and a 12-volt adapter to power the solenoid valve. With these parts we build a circuit like the one in the figure below:

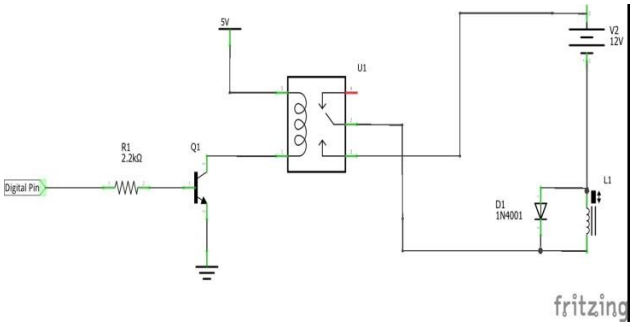


Fig. 1 Solenoid valve drive circuit

When the digital pin is switched on, the transistor turns on and current passes through the relay coil. As current passes through the coil, the switch moves toward the normally open contact, the circuit of the adapter and the solenoid valve is closed, and the solenoid valve turns on.

To read the moisture sensor data we also use one of the ESP32's analog pins. The overall connections to the ESP32 are as shown in Fig. 2.

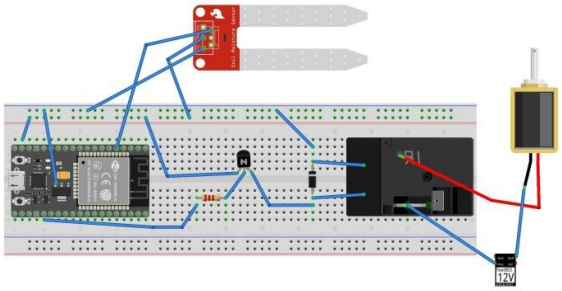


Fig. 2 How the sensor and solenoid valve are connected to the ESP

2-3 Building the Web Page

To build the web page we use Node-RED. First, to open the Node-RED programming environment, we enter the address Raspberrpi_IP:1880/ in the browser (Raspberrpi_IP is the IP address of our Raspberry Pi).

To connect Node-RED with the MQTT broker, we add the relevant nodes to our program. Their internal settings are as in Fig. 3. The broker name is the Raspberry Pi's IP address, and for each message we enter its related topic. These topics are the ones we wrote in the ESP code. For example, to receive the moisture sensor data of Esp1, we write tree1/humidity in the topic field. In this case Node-RED receives from the broker the information that Esp1 has

published.

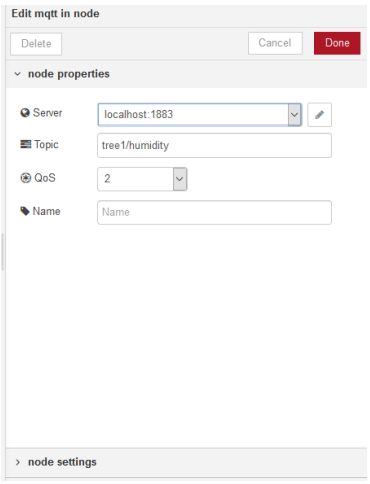


Fig. 3 MQTT settings

After that we add a gauge to our program, and the sensor data is displayed with it. See its settings in Fig. 4.

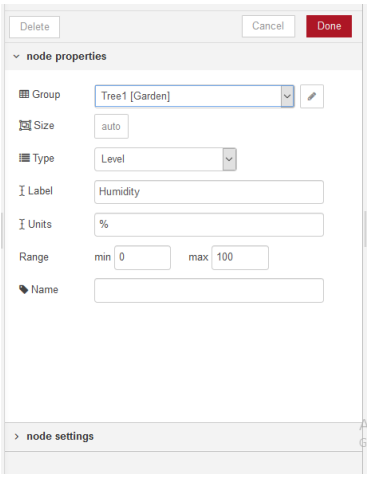


Fig. 4 Gauge settings

We also add a switch to the program, which in fact passes the on and off messages to the mqtt node, and the mqtt node publishes this message to the broker. In this way we can turn the solenoid valve off and on with this switch, because the ESPs receive this message from the broker and turn the actuator off and on accordingly. We add these nodes to the program for all of the ESPs. See the overall program in Fig. 5.

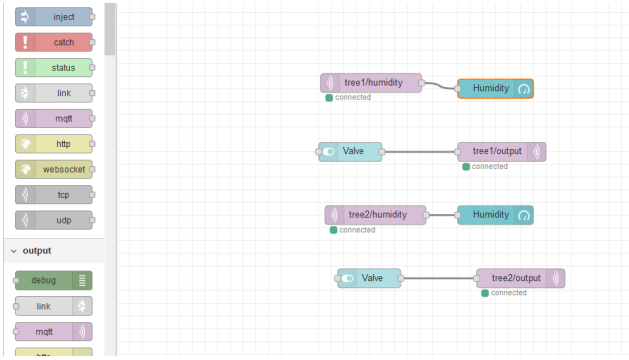


Fig. 5 Overall program in the Node-RED environment

Now we click the Deploy button. In this way the web page for the program is built, and we can access it through the address (raspberry_ip:1880/ui).

The web page can be seen in Fig. 6. In this way, as long as the Raspberry Pi and the ESPs are connected to the Internet whose username and password we have entered in the ESP code, we can access the sensors and the actuators attached to them through this web page.

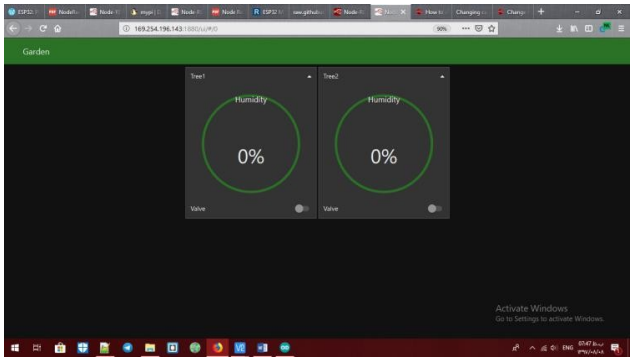


Fig. 6 Screenshot of the web page built with Node-RED

2-4 Accessing the Web Page from Outside the Local Network

There is one point to note. To access the web page we must be connected to the same Internet network that the Raspberry Pi is connected to. That is, if we are somewhere else, outside the house or garden, we cannot access this web page. This is because our router's firewall does not allow it. One way to access the web page is to change the router settings. Another way is to use the Dataplicity service and the nginx server. Dataplicity is a service with which we can access the Raspberry Pi's terminal from anywhere. The Dataplicity service also lets us reach a website hosted on our Raspberry Pi through a public link. Using the nginx server, and installing it on the Raspberry Pi, we then make that public link point to the Node-RED page.

In this way, wherever we enter the public link, the project's web page opens for us, and we will have access to the web page from outside the local network.

2-5 General Design

Now that we have explained how the system is implemented, we want to describe the overall operation of the system. Look at Fig. 7. The system works like this: from anywhere we have Internet access we connect to the web page, and we can then see the values of the moisture sensors. These values come about as follows. The WiFi module has published its sensor value to the broker under a specific topic, and the web page, in order to receive messages under that same topic, has subscribed to the broker. The sensor value is therefore passed on to the web page.

We can also turn the solenoid valves off and on using the switch on the web page. This happens as follows. When we turn the switch on, the web page sends the message “on” to the broker under a specific topic. Meanwhile, one of the

ESPs has requested messages under that same topic from the broker. The broker therefore sends “on” to that ESP, and that ESP, based on the message and an if condition in its code, turns on the pin connected to the solenoid valve.

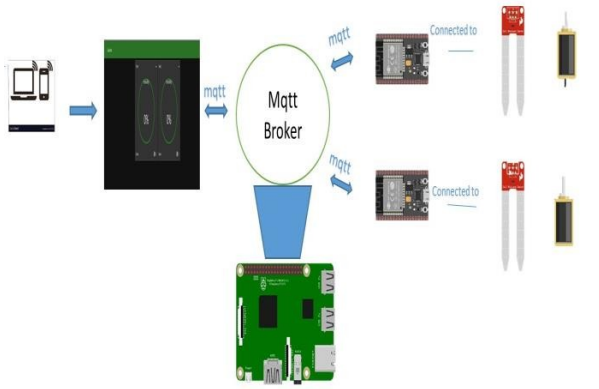


Fig. 7 General design of the project

3- Conclusion

In this way we were able to build an irrigation system based on the Internet of Things. From anywhere we have Internet access, we can read the values of the moisture sensors, turn the solenoid valves off and on, and control the irrigation process. There are also suggestions for extending the work. We can easily extend this system to larger scales and to more sensors. For example, we could use sensors that measure the minerals in the soil and display those values on the web page.

One way to increase the security of the system is to require a username and password to enter the web page, which is easily possible by changing the Node-RED settings.

Since the solenoid valves and WiFi modules are located outdoors, we can use solar cells to power them.

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